

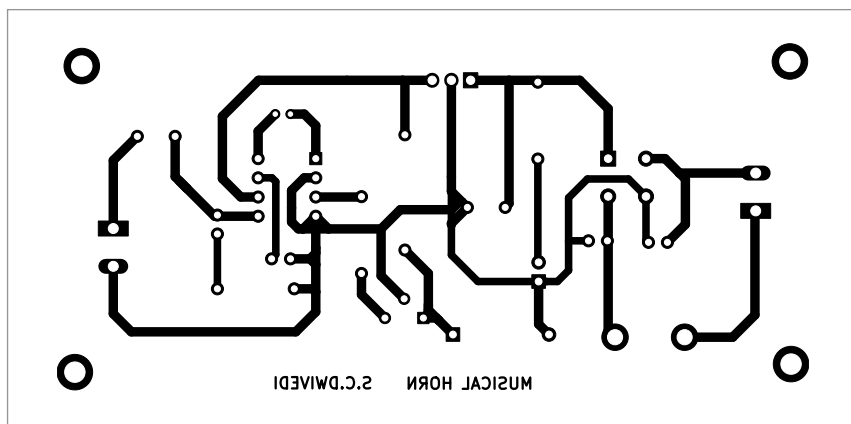


Fig. 3: Actual-size PCB of musical horn

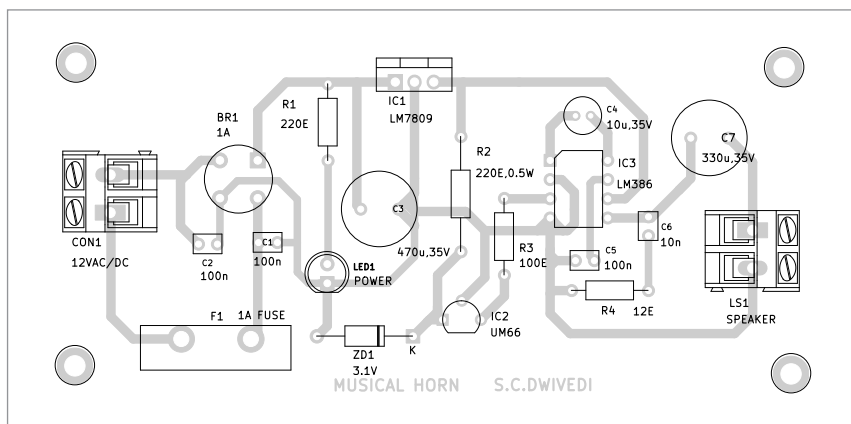


Fig. 4: Component layout of the PCB

UM66T05 – Home sweet home
(a nice melody)

UM66T06 – Let me call you
Sweetheart

UM66T08 – Happy Birthday to
you

The UM66 melody tone generator IC used in the circuit may be changed to any other in the series to change the melody.

This musical horn circuit can work on 12V DC as well as 12V AC. If you want to operate it on AC mains, use a 230V AC primary to 12V, 500mA secondary transformer to step down the AC mains voltage to 12V AC. The stepped-down AC voltage may be connected directly across CON1 in the circuit.

The 12V AC voltage is rectified using bridge rectifier BR1 and filtered by capacitors C1 and C2 to remove any ripples in the voltage.

The rectified and filtered voltage is then regulated to 9V by regulator IC 7809. In case you want to use 12V DC, connect it directly across CON1.

The LM386 is a power amplifier designed for use in low-voltage consumer applications. Its gain is internally set to 20, but the addition of an external resistor and capacitor between pins 1 and 8 will increase the gain to any value up to 200. The amplifier's inputs are ground referenced while the output automatically biases to one-half the supply voltage.

The operating voltage of the circuit is 12V AC or DC. The UM66 works between 1.5V and 4.5V DC, so a Zener diode has been used here for providing 3.1V to UM66. Power amplifier LM386 works on 4V to 18V; here it uses 9V generated by IC 7809 for a sufficiently loud sound.

PARTS LIST

Semiconductors:

- IC1 - 7809, 9V voltage regulator
- IC2 - UM66 melody generator
- IC3 - LM386 low-power audio amplifier
- BR1 - 1A bridge rectifier
- ZD1 - 3.1V, 0.5W Zener diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 220-ohm
- R2 - 220-ohm, 0.5-watt
- R3 - 100-ohm
- R4 - 12-ohm

Capacitors:

- C1, C2, C5 - 100nF ceramic disk
- C3 - 470 μ F, 35V electrolytic
- C4 - 10 μ F, 35V electrolytic
- C6 - 10nF ceramic disk
- C7 - 330 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- LS1 - 8-ohm, 0.5-watt loudspeaker
- BATT.1 - 12V battery (or 12V AC)
- F1 - 1A fuse with holder

The output of IC UM66 is given to input of LM386 for amplification of the melody generated by UM66. The LM386 has a 10 μ F capacitor connected between its pins 1 and 8, which is used for the gain of the amplifier. A push button/momentary switch (not shown in circuit diagram) is used to turn on the circuit, thereby producing a loud horn sound.

Working of the circuit is simple. After connecting 12V AC/DC across CON1, the circuit is ready for use. As soon as power is switched on, LED1 glows and at the same time the speaker produces a musical sound.

An actual-size, single-side PCB layout for the musical horn is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 in front of box and the speaker on the rear side. The circuit is now ready for use. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at: <https://www.electronicsforu.com/videos-slideshows/live-diy-musical-horn>

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